October 27, 2026

Letter of Support for the ICRA27 Proposed Workshop "Who Coordinates Whom? Adaptive and Safe Coordination of Heterogeneous Multi-Robot Teams in Human-Centered Dynamic Environments"

To Whom It May Concern:

As one of the Co-Chairs of the IEEE Robotics and Automation (RAS)'s Technical Committee (TC) on Human-Robot Interaction, I would like to support the initiative of Silvia Proia (University of Modena and Reggio Emilia, Italy), Esteban Restrepo (IRISA, CNRS, France), Graziana Cavone (Roma Tre University, Italy), Martina Lippi (Roma Tre University, Italy), Keisuke Okumura (National Institute of Advanced Industrial Science and Technology AIST, Japan), Pratap Tokekar (University of Maryland and Amazon Robotics, USA), Changjoo Nam (Sogang University, South Korea), Antonio Marino (University of Cambridge, UK), Lorenzo Sabattini (University of Modena and Reggio Emilia, Italy), concerning their submission of the workshop proposal "Who Coordinates Whom? Adaptive and Safe Coordination of Heterogeneous Multi-Robot Teams in Human-Centered Dynamic Environments" to the 2027 IEEE International Conference on Robotics and Automation (ICRA 2027) as its objectives and topics of interest are within the scopes of the TC.

This workshop addresses a central challenge in human-centered robotics: how heterogeneous multi-robot teams can adapt their collective decisions and actions to evolving human behavior while ensuring safety and maintaining task efficiency. Bringing together researchers and industry experts from multi-robot systems, human-robot interaction, artificial intelligence, planning, and control, the event will explore four interconnected themes: human behavior understanding and safe interaction, human-aware coordination under uncertainty, AI-driven predictive and adaptive coordination, and real-world deployment.

It directly aligns with the scope of the IEEE RAS TC on Human-Robot Interaction by addressing safe, adaptive, and effective interaction between humans and heterogeneous multi-robot teams in dynamic environments. Particular emphasis is placed on human behavior understanding and prediction, human-aware planning, social navigation, and safe human-robot collaboration. Moving beyond individual human-robot interactions, it investigates how human intentions, actions, and preferences can inform collective robot decision-making, task allocation, and coordination. A workshop at ICRA27 on this topic will provide an opportunity for researchers in the field to share their ideas and present the latest research and application developments, by encouraging discussions about the on-going and next challenges to accomplish.

Your sincerely, on behalf of the Chairs of the IEEE RAS TC on Human-Robot Interaction

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